

The Adoption of Automated Building Code Compliance Checking Systems in the Architecture, Engineering, and Construction Industry

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Abstract — Most Automated Compliance Checking (ACC) systems for building codes currently in use cannot fully translate building-code requirements into forms that can be processed by computers, especially when such requirements have complex semantic and syntactic structure hierarchies. Modern information extraction and transformation rules-based ACC approaches, capable of handling complicated requirements, are rigid when used with various regulatory documents. To assist the ACC systems' downstream responsibilities, like information matching and compliance reasoning, a flexible approach to automatically processing and interpreting requirements must therefore be developed.

This study is based on previous relevant research and, despite being integrated based on the authors' knowledge from various based on disciplines, draws on previous research.

Within the scope of this study, we looked into the history of issuing Building Permits (BPs) and the required process, technology, and rules to get the ball rolling on automating the process of checking compliance with building codes. Finally, based on the findings of the research, we defined the required steps to start automating the building code compliance checking system in the Kingdom of Bahrain.

Keywords—digital building permit, automation of compliance checking, ifcOWL ontology, BIM-based code compliance, AEC, Kingdom of Bahrain.

I. INTRODUCTION

The application and checking process for BPs in many countries, including the Kingdom of Bahrain, is still considered a subjective, manual, time-consuming task that may lead to uncertainty and inconsistent assessments due to human error.

In the industry of Architecture, Engineering, and Construction (AEC) in the Kingdom of Bahrain, there are many standards and codes to consider during the early design stage of any building, most importantly the design compliance with building regulations to ensure the building's stability and usability.

Many Architectural and Engineering (AE) firms in the Kingdom of Bahrain fail to comply with local building regulations when submitting their first design package for checking to obtain a Building Permit (BP) from the auditing AE office and ultimately the competent authority, the municipality. Repeated failure to comply with building regulations has a detrimental impact on several areas, including the forecasted project schedule and, as a result, the cost and quality of the project, as well as the professionalism and credibility of the firm.

In recent years, there has been a global push toward digitalization, which is now made possible by technological advancements in data management and processing in the AEC industry. Digitalization in AEC has also been a major priority for public administrations because it opens up new opportunities for innovation, particularly in the areas of issuing BPs and compliance checking. In this sense, the legislative and policy context, knowledge gaps, technological uptake, and other related concerns all provide important challenges [1].

In addition, a network of researchers and stakeholders known as the European Network for Digital Building Permits (EUnet4DBP) was formed in recent years. This network aims at establishing a standard approach for the digitalization of the BP issuance process.

The mission of EUnet4DBP is to accelerate the BP process's digital revolution. In March 2021, the EUnet4DBP conducted a workshop along with EuroSDR (a not-for-profit organization linking national mapping and cadastral agencies with research institutes, universities, and companies in Europe) and buildingSMART (a worldwide industry body driving the digital transformation of the industry of built assets). The workshop's primary goal was to present some pertinent works on the topic and use them as a forum for discussion for an international, intersectoral, and multidisciplinary audience to raise awareness about the state of the art, pertinent pilot studies, and leading experiments, as well as common challenges and the current state of practice. The workshop was organized by the EU-BIM Task Group (formed to deliver a pan-European approach to best practices in Building Information Modeling (BIM)). The foundation for upcoming innovations is anticipated to be set by such a knowledge exchange [1].

Despite the benefits, it brings to the built environment and the increasing maturity of asset information models, automated compliance checking has not yet been widely adopted. However, as asset information models mature and BIM is adopted, automation of compliance checks becomes viable [3]. Some countries have already begun to take steps to automate the process of checking AE drawings before issuing BPs; examples were discussed in the EUnet4DBP workshop. The Italian government has started working on a plan to push public administrations toward digitalization. In this regard, a Ministerial Decree (D.M. 560/2017) was published in 2017 to control the required deployment of BIM methodologies in public procurement beginning on January 1, 2019, and issuing BPs is being impacted by this transformation. The

State of Geneva introduced the "AC Demat" digital platform for BPs in 2019, and it now delivers over 80% of the BPs completely dematerialized using online forms and 2D pdf blueprints. A BIM and geographic information (GEOBIM) strategy is being developed. Concerning the usage of BIM for BP, a guide was created in Portugal with the primary objective of increasing municipal awareness of the issue of BIM and providing guidance for getting ready for its implementation.

Another example was demonstrated during the EUnet4DBP workshop by Dubai Municipality in the United Arab Emirates. The BIM Roadmap was launched in January 2020. The geographic information system (GIS) center in Dubai Municipality oversees project management. The initiative intends to raise the caliber of GIS and BP services. The project's main output objectives include: (1) drawing a BIM roadmap for Dubai's AEC industry; (2) unification of BIM submission standards for BP applications; (3) a BIM e-submission platform with an automated code compliance checking engine; and (4) building model migration to GIS. The Swedish model shares similar characteristics concerning the consideration of the overlap between the BIM and GIS domains in the AEC industry. The model's consideration of data exchange protocols enables the integration of BIM with geodata for urban planning and 3D city modeling, which is particularly apparent when handling BP data [2].

Furthermore, during the workshop, Hjelseth [3] referred to two systems adopted by two different countries: the "ByggSk" (Building-Application) solution, which was created by the Norwegian Building Authority in 2007, and which is a web-based method for confirming that application forms for particular kinds of applications have been filled out correctly. Also, Singapore's Construction & Real Estate Network (CORENET) e-information system, which dates back to 1995, is a one-stop shop that enables building professionals to submit their applications to get BP electronically from 16 government regulatory authorities from anywhere at any time. In 2018, the CORENET X project was launched to transform the regulatory approval process for building work by leveraging technology such as BIM and automation. CORENET X is envisioned as the future regulatory approvals ecosystem, reinventing how the government and businesses interact as a one-stop digital storefront. It will go live by the end of 2023 and uses an agile method [4].

Therefore, it stands to reason that these systems would have been employed in other countries, yet they were not.

The Kingdom of Bahrain's e-government launched a BP portal in 2018 named Benayat, the Arabic word for building, to issue permits for all building types. The portal enables AE firms to apply for BPs, check the progress of their applications, engage with the relevant governmental bodies about BPs, pay required fees associated with BPs, and apply for additional building-related services. Furthermore, Benayat allows applicants to collect all of the data required to generate AE drawings and other paperwork required to apply for a BP [5]. Even though the portal lets documents and AE drawings be sent electronically in PDF, DXF and other formats, the benefits of increasing the level of automation have not been fully realized yet.

As a result, the subject of this study is to conduct a systematic review, intending to establish a roadmap to

gradually transform the current BP system into a fully automated one capable of checking AE drawings' compliance with specific building regulations, with a focus on the system used in the Kingdom of Bahrain. This paper attempts to analyze the published literature over the last five to seven years, with a focus on the most recent ones, in terms of the technological and administrative issues connected with components of building permit system automation.

Although it has attracted a lot of interest from both academics and businesses, the issue of BP digitization and automation is still quite young. Due to the difficulty posed by the numerous underlying problems, such solutions often focus on a small number of discrete aspects of the more complicated process and associated problems. To understand what's going on in the area, it's important to make a clear overview of the current efforts using a common framework. This will show how complicated the problem is.

The five sections of this study are as follows: The development of BP issuance is discussed in Section 2. The research methodology used to produce this study is outlined in Section 3. The discussions, as well as the limitations relating to the advancement of BP complete automation, are given and further discussed in Section 4 and Section 5 concludes the paper.

II. Development of Building Permit Issuing

Historically, the goal of ACC research has been to provide accurate compliance checking of building projects against relevant building regulations with time and cost-effective computational support. The representation of building codes in computational format, the definition of building model views, and the compliance checking algorithms and reporting have received the majority of attention in research into constructing ACC systems. Building codes must be retrieved by ACC systems from relevant authorities, and submitted building projects must be checked for compliance. Computer-based models of appropriate building codes and building designs are a key requirement for compliance-checking systems. Even though it is now regarded as insufficient, improvements in BIM technologies have ultimately developed a consistent representation for the design of different building types, yet a standard representation for building codes in a machine-readable format is required [6].

Around the 1980s, the field of AEC began to develop BIMs. These tools, which are an outgrowth of Computer-Aided Design (CAD) tools, have gained popularity over the past 20 years. The usage of the same native format generated by BIM software could be one way to achieve potential. However, from the perspective of developers and users, using open standard formats—in particular, for BIM, the Industry Foundation Classes (IFC), which is a CAD data exchange standard designed to describe data for the AEC industry and is also defined as a conceptual model schema in a neutral file format to assist building-related information exchange in the AEC industry would be a more desirable and practical approach.

BIMs are represented by IFC, a comprehensive, extensive, and complicated open standard data model. buildingSMART, the standardization group, is working hard to create and maintain this standard. Some challenges appear when attempting to use IFC models from practice for automatic analysis [7], [8].

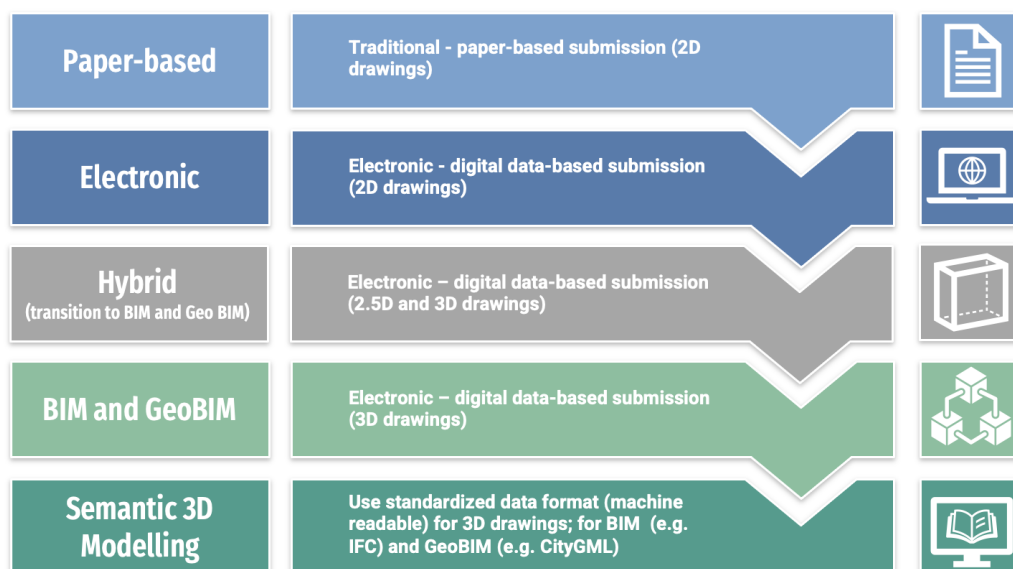


Fig. 1. Development of building permit issuing – Source: adapted from Noardo et al., [7]

The limited software vendor acceptance of compliance processes and the distributed development of ad-hoc methods for monitoring and achieving compliance with rules and requirements at various phases of the construction life cycle constitute the present state of the art in this field. These makeshift solutions are not scalable, can't be moved from one building to another, and are difficult for non-technical people to operate. This is shown by the fact that continuous assessment, which is the process of checking an asset's compliance with regulations regularly over a set period, is rarely used in operational buildings. This demonstrates that constructed assets are not managed in a systematic manner [9].

According to several studies, the process of going digital is ongoing in various industries, including AEC. To help with the structure and management of digital information about AEC projects, the ISO 19650 standards were created. This would make several benefits possible, including the improvement of communication and information flow as well as information sharing and reuse. Improvements in resource utilization and monitoring at a more central level, better design and execution of building projects, and other positive benefits would all follow as a direct result. The digitalization of the BP process toward automation is gaining interest. This change is intended to speed up compliance checking with optimized resources, with benefits to the quality of both the city and the built environment [11].

Additionally, the BP is issued in a very complicated process that involves numerous processes and parties and the use of numerous pieces of data. In addition, the techniques and data that are involved in the practice are extremely diverse and, in most situations, relatively informal; for instance, choices depend on the local knowledge and skill of individuals. It's crucial to address the many components inside a common framework. The ACC has been studied for almost 70 years and has recently become one of the most well-liked research topics in the field of construction. Meanwhile, as commercial software, semantics, and cognitive technologies advance, ACC becomes increasingly automated. The four stages of ACC can be broadly divided into: (1) building model preparation; (2) rule interpretation; (3) rule execution; and (4) the reporting of the checking outcomes.

Building model inputs from IFC documents were often considered throughout the step of building model preparation. During the building model preparation stage, IFC documents are usually turned into Resource Description Framework (RDF) standard models for certain semantic ACC applications [12].

Fig. 1., which is adapted from [7], best describes the development of BP issuing from a traditionally analog, paper-based process to a fully automated, model-based setting.

III. METHODOLOGY

The literature review that is being discussed in this study takes into account both technical and scientific contributions. These were retrieved and analyzed using the research approach that is being discussed in Item A of this section. It began with the gathering of pertinent contributions classified by the aforementioned keywords. The scope of the task of examining the selected papers is to analyze trends in the fields of Digital Building Permits (DBP) and AEC to determine the degree to which the scientific community is interested in a global scale. The scientific contributions that were presented during the EUnet4DBP workshop were also considered along with the papers that were selected. The three pillars of the EUnet4DBP were used as a guide to help extract pertinent information from the selected papers, as is outlined in Item B of this section. In the final step, the results were summarized and discussed so that an appropriate conclusion could be drawn.

A. Selected Papers

To study DBP and AEC-related research completed globally over the last five to seven years, with a focus on the most current. The Connected Papers tool was utilized to find related open-access papers. This database was chosen because it provides a visual overview of the selected research field and allows the user to locate at a glance which publications are most popular, as well as the numerous dynamics between areas of research. As a result, it allows the user to ensure that no important material is overlooked. The advanced search tool, which takes into account the title, abstract, and keyword, was utilized to locate relevant technical

and scientific contributions. The search included journal articles, academic theses, conference proceedings, etc.

The authors evaluated the quality and originality of the found papers before applying another filter to limit the final selection to quality papers published by reputable publishers such as SAGE, MDPI, IEEE, Elsevier, and others. The familiarity of the authors with AEC and business information systems and management and the recently conducted workshop by EUnet4DBP led to further integration.

B. Extracted Information from the Reviewed Papers

The information in the selected research papers was analyzed based on the following criteria, guided by the three pillars of EUnet4DBP: process, technology, rules, and requirements.

IV. DISCUSSIONS

The selected papers are reviewed against the three pillars that the EUnet4DBP activities are built on, and that is done to extract the relevant information for DPB and AEC:

A. Process

As the status quo changes, it is essential to reconsider how to organize and handle data, making use of the abundance of available digital resources and altering the duties of various stakeholders accordingly. This necessitates not just new training for operators but also agreements with stakeholders to assure their active participation in this transition.

Many researchers pointed to the challenges associated with adopting new workflows. Fronk [13] discussed how a common data environment could improve stakeholder communications by cutting down on time wasted and lowering the chance of data loss. In Italy, beginning in 2019, all complex construction projects costing more than 100 million euros must use the BIM approach. BIM will be necessary for all categories of public works starting in 2025. The public administration is expected to publish common data-generating standards and release the necessary documentation as part of buildingSMART's implementation of open standards as IFC.

The public would be able to monitor and manipulate their data in a similar way, which would help the designers keep an eye on these elements.

Noardo, Guler, et al. [7] agree with Fronk that obtaining a BP entails a variety of steps, as well as the engagement of a large number of stakeholders and the use of a large amount of information. Many countries continue to deal with this type of information in analog forms (or, at best, PDF documents). The shift from existing analog workflows to new digital workflows would undoubtedly necessitate an in-depth investigation on its

own. Fig. 2, adopted from Noardo, Guler, et al., [7], depicts the workflow phases 1–8 that were considered and how they were assigned to the participants in the DBP process. It was observed that the most significant gaps are in the use of geoinformation (Step 2), notably in relation to BIM and reciprocal conversions (Step 6). More GeoBIM research has been developed in recent years; nevertheless, these studies are not directly related to DBP. There are currently very few studies that focus on the pre-consultation phase (Step 3), most likely because it requires the preliminary settlement of several diverse difficulties. There are currently very few studies that focus on the pre-consultation phase (Step 3), most likely because it requires the preliminary settlement of several diverse difficulties.

In another study by Noardo et al. [14], which focuses on integrating GeoBIM approaches to develop the issuing of building permits process, the authors emphasized that it would be advantageous for both parties if the designer and municipal organizations that provide building licenses were to interpret constraints with more objectivity. According to the authors, by implementing GeoBIM, the municipality will be able to use the rich data offered by the design and correctly georeference the 3D city models effectively and objectively.

The 3D city model will be kept up-to-date with the same checked data. Instead of using numerous separate different pieces of data that would be lost after the process, one building model and a 3D city model would be used in a seamless workflow, and this would result in fewer errors.

The same study demonstrated how combining geoinformation with BIM (GeoBIM) could assist in resolving several city-related issues. The outcomes demonstrate a standardized process that automates planning permits using GeoBIM data. The authors continued by explaining that a GeoBIM method offers substantial advantages over the current situation, which is based on 2D cross-section drawings and a 2-2.5D scenario plan in most countries for the BP procedure. With GeoBIM, many building regulations might be automated. It was demonstrated that building heights, including roof shapes, major viewing angles, etc., could be assessed using GeoBIM in a Swedish case study.

The transformation of national governments into digital entities is one of the foremost concerns of all nations. For instance, the European Union is now implementing projects to enhance countries' e-government and digital public service delivery capacities. As a direct result of this, it was officially proposed for the first time to use BIMs for BP procedures. Recently, it was revealed that the European Commission would build a framework for a digital permit system in the AEC industry across the European Union. Several countries are now working to switch from two-dimensional (2D) to three-dimensional (3D) cadasters. This is due to the inability of a 2D demarcation of legal rights to adequately reflect the

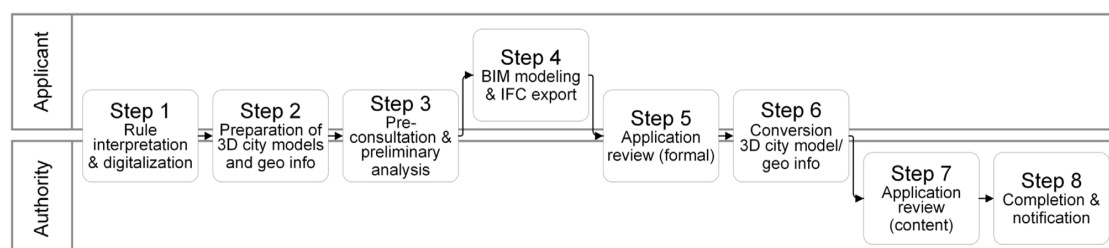


Fig. 2. Overview of considered workflow steps assigned to involved parties – Source: adopted from Noardo, Guler, et al., [7]

reality of property ownership in multipartite buildings. Since some nations execute more than 100,000 BPs a year, the transition to DBPs requires a substantial amount of effort. Administrative structure and country size are particularly effective elements in the digitalization of the BP application process [15].

The new Dutch Build Quality Assurance Act increases contractor duties to improve building processes and product quality. To achieve this, a contractor must evaluate whether engineering offices' designs meet building code standards and ensure that as the design is realized and adjusted, the finished building meets building code standards. The contractor must conduct independent verification [16].

As a result of the Directive 2014/24/EU of the European Parliament and the Council on public procurement [10], which strongly encourages the use of BIM for public projects, it is often driven by national BIM strategies in which governments define rules and use cases where BIM must be used for documentation and management tasks in the AEC industry. Examples include the UK, the Netherlands, and the majority of other European countries. In various nations, such as Slovenia, Spain, Italy, France, and Germany, a project to apply BIM for public buildings by 2022 or later got underway in 2018. There may be no official regulations in some other countries, but authorities, such as Swedish transport administration, may choose to impose BIM in construction projects. Moreover, there are ongoing projects [14].

B. Technology

In the field of technology, incredible advances have been made, allowing for the pursuit of various new areas of effort. Several innovative tools and applications, as well as some excellent examples of their use, were exhibited during the EUnet4DBP workshop. The fundamental barrier is their adoption into existing practice, although the technology solutions that are currently accessible offer a lot of promise.

Hjelseth [3] talked about how the digital solution could range from making decisions automatically without human help to helping people make decisions by giving them relevant information (full application).

The Semantic information is expanded by converting ifcOWL to railway code ontology, this technique ensures the hierarchical connection of code ontology classes with good scalability. By dividing ontology development into the low-coupling processes of extension and conversion, the need for subject expertise among researchers is decreased. To create a clause-checking code ontology, a practical specification is finally selected. To provide traceable, transparent code-checking applications, a rigorous, systematic, logic-based approach to qualitative performance goals and design requirements is required. To address implicit, imprecise, and flawed human knowledge about building-specific rules and restrictions, they propose a specialized spatial reasoner [17].

The integration of explicit knowledge representations and heterogeneous information from various sources is made possible by semantic web technologies. Knowledge is modeled and transferred into a form that both computers and people can comprehend using an ontology, a crucial component of the semantic web. Ontology can assess whether or not a building complies with the fire code, for example, or other regulations. To communicate specification semantics,

code ontologies are employed in compliance verification. The model data on the code ontology is frequently checked using rules. Before applying a compliance check, it is important to assess the model's information integrity [17]. Other researchers [18] presented a workflow for multiscale information integration based on semantic modeling that is based on restructuring data calls after exporting BIM data in IFC and then following the CityGML schema (which defines a conceptual model and exchange format for the representation, storage, and exchange of virtual 3D city models). The output is used in a variety of applications and is kept in an urban database. Roles and responsibilities must be defined, and the process must be standardized. Understanding data structure is crucial for digital information sharing to prevent data loss and exploitation. Although the procedure is intricate, it may be scaled back and repeated.

Another integrated approach that was talked about was the BIM bots system, which is made up of different "Robots". The future of code checking would use digital robotization—an ecosystem of autonomous BIM cloud applications dubbed BIM bots. A major part of actual testing should be supported by automated technologies. The controller can then focus on the building's context and practical difficulties. Using open, internationally acknowledged buildingSMART and BIM standards, including IFC, makes communication between information carriers (aspect models) possible. TNO (an independent Dutch research organization) developed the BIM bots technology recently. "Robots" (BIM bots, see Fig. 3.) can autonomously read data from a BIM model, analyze it, and deliver the results to the user. BIM bots parse IFC and relate external product properties. Smart procedures can estimate the measurements of walls, floors, doors, height disparities, and rooms, and technology is being developed to link IFC attributes with supplier product databases or generic material databases. Machine learning is used, although interpreting IFC data is another issue. Simple things like composite wall thickness can be a challenge without a BIM model [16].

The use of software like Solibri, from the same-named firm, which deals with model and code checking, was also discussed during the EUnet4DBP workshop. With the help of Solibri Office, a user can automatically launch a set of regulatory verification rules and store the results of the analysis in reports in the conventional formats of PDF and XLS, or directly in 3D format, etc. In Solibri, there are several fundamental criteria (such as the minimum distance between objects and the minimum object dimensions) that can be combined to apply any particular code checking. Any differences between the model and the rules in effect will be

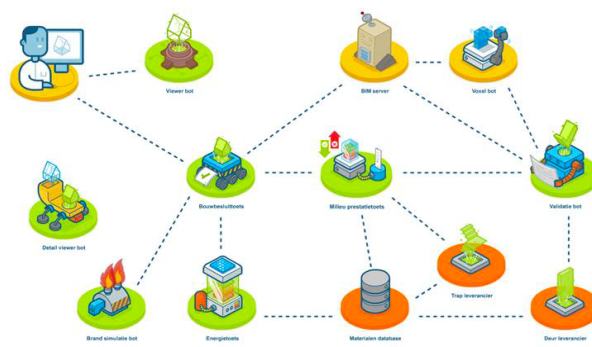


Fig. 3. BIM Bots system with various “Robots” – Source: adopted from Straalen et al., [16]

automatically organized into a report after the checks have been completed. All extracted reports will be shown in 3D graphics along with the problematic components and technical notes [19].

Other BIM-based technologies can also be considered for automating building code compliance checking systems. Today, the AE design process incorporates both augmented reality (AR) and virtual reality (VR), as BIM and AR technologies can enhance decision-making and enable real-time access to information [20], and AR can overlay and compare BIM design to what was constructed. The benefits of using AR capabilities include monitoring code and compliance [21]. Efforts are made by researchers to adapt AR/VR maturity models to BIM. Researchers from France discussed that the challenge of increasing the efficiency of the various phases of a building's or infrastructure's projects, and hence its life cycle, necessitates consideration of cutting-edge technologies such as AR and VR systems, which seek to virtualize or augment the content of BIM in real time to support continuous improvement [22]. Researchers from South Africa found that augmented reality can improve construction productivity through its applications in design information, monitoring and controlling, etc. [23]. Other researchers from Switzerland looked into how VR enabled building-data management through the integration of an IFC-BIM model and an Internet of Things (IoT)-based building management system [24].

C. Rules and Requirements

The utilization of open standards, most notably IFC, is universally regarded as the most efficient approach going forward. In addition, it is the responsibility of the designers to ensure that all standards and specifications are adhered to for their models to function properly within automatic tools. Various methods, such as Excel sheet requirements or particular IFC model view definitions, are among the options that have been suggested by many researchers. Researchers from Portugal analyzed four Portuguese codes and offered a good example that can describe the rules' impact on the automation of the construction permit. This example can describe how the rules can affect automation. These codes are written in a non-machine-readable language, and they specify the rules that apply to the inside as well as the exterior of many different types of buildings. Note that not all of the rules from the codes are automatically verifiable because a new business endeavor utilizing an algorithm-based technique is now in the process of being developed. The method of employing an algorithm to geometrically verify which internal spaces are contained within each building fraction was examined by the authors. A code was used in a 2D environment to verify that every point in each space was located inside the polygon that belonged to the fraction. The area of each interior space was evaluated, and an algorithm was used to determine whether or not it met the prerequisite for having a certain minimum area based on the fraction typology and the room name. In Fig. 4., adopted from [25], some instances of rules are provided to explain the methodology that was followed.

Other examples of rules and requirements are guidelines, common data models, shared vocabularies, etc. Examples of work aimed at interpreting rules as algorithms include collaboration with checkers in municipalities, parameters defined in spreadsheets, programming languages for storing regulations, etc. Specifying the types of geometry to be

utilized, the necessary objects, how semantics should be employed, the appropriate level of abstraction, and so forth are the result of a combination of implementation requirements and rules and regulations [7].

Other researchers said that the rules and requirements were the most important thing that affected the change to a fully automated BP issuing model. Researchers from Germany conducted a project about BIM integration. The BIM model integrates other sources and technologies. When a project is planned, the local rules on land use and building layout must be evaluated. Arrangement plans comprise the physical layout of the construction site and design limits such as building area, height, or the number of floors. These plans should be available and exchanged in a single data format. Note that, many parameters, especially quantities, can be derived from the BIM model. Only fully modeled elements with all required properties assigned will allow for automated processing of the model. As a result, exchange criteria need to be specified and taken into account. Many of them refer to the existence and proper format of properties, but it's also crucial to take into account the geometric needs of the geolocation to keep them in sync with design goals. These details are typically recorded using open standards like property- and quantity-set definitions or IFC template properties, in addition to the consolidation of modeling principles for particular and local application processes [26].

Researchers from Denmark completed a study on a logic-based method for verifying performance-based codes in the area of digital building permit procedures. The researchers expressed that their goal is to retain, transfer, and exchange high-level requirements and constraints by linking a building's physical manifestation with its architectural rationales. This will enable a flexible, human-centered, and ecologically responsible approach to building compliance audits within the scope of EUnet4DBP, with a major focus on the guiding principles and objectives of performance-based building regulations [27].

In terms of ontology application, the major requirement for ontology modeling in engineering sectors is the capability to define (ordered) collections of idea instances, such as wall material layers, work sequences, or fundamental things like coordinates. On the other hand, due to the nature of Description Logic (DL)-based ontologies, concepts and the instances of those concepts are regarded the same way as sets and the members of those sets. Additionally, there are very few or no built-in processes for ordered structures. Two potential solutions to this problem are the construction of ordered collection-like structures on an upper ontology level using OWL-DL constructs and the use of lower-level RDF constructs without going outside the DL complexity box into OWL-Full. The authors have successfully applied both of these potential solutions to the ifcOWL case [28].

Other researchers from the Netherlands discussed the use of GeoBIM techniques for the digitalization of building permit checks and the importance of coming up with and settling on a set of standards that would standardize how a particular scenario ought to be modeled in IFC. They went on to say that the history of issues related to the interchange of data was far more extensive in the GIS domain compared to the BIM sector [29].

Code/Article	Rules	Rules and methodology used
RGEU/ Article 66º §1	The housing compartments may not be in number and area lower than those indicated in the code	All the IfcSpace must be named correctly (fractions and interior spaces). The interior spaces could be classified as SINGLE ROOM, DOUBLE ROOM, COUPLES ROOM, LIVING ROOM, LAUNDRY ROOM and KITCHEN. The number of rooms and typology of the fraction will be verified based on the area of the interior spaces
PDM/Article 38º §1	The full occupation of the building is not allowed, with constructions, even if in the basement, being the maximum deployment area, 75% of that area, not counting the areas of service to the domain for the purpose of calculating this percentage	The IfcSpace of both the implementation area and of the land parcel are needed. The percentage is calculated and checked for compliance
Urbanization Plan for Avenida da República/ Article 24º	§1: The maximum depth of construction at floor level 1 of buildings with two fronts cannot exceed 35m §2: The maximum depth of construction of the levels above the floor level 1 cannot exceed 17,5m	The IfcBuildingStorey must be named correctly and the distances between the front and the back exterior walls will be measured

Fig. 4. The methodology used to verify rules from the local – Source: adopted from Magalhães et al., [25]

For modelers to supply adequate data for automated tools, several guidelines must be provided. Simple data export to the IFC format is insufficient; first, one must ensure that data is preserved and not corrupted during conversion from proprietary program formats. Along with the proper grouping of floors and depiction of spaces and apartments, consistent semantics and a thorough filling up of the information required for the use case are also crucial. The important factors identified during the models' assessment and found to be pertinent for implementation are provided as guidelines, mostly for building information modelers, with a few recommendations also for geoinformation providers and city planners [30].

V. CONCLUSION

When it comes to the exchange of data regarding DBP, interoperability between different stakeholders is critical. The most successful technique, it is generally acknowledged, is to incorporate the standards into the processes. At this point, it's important to note that the standards can be used by countries as a basis for creating national standards that meet the specific requirements set by the DBP.

Furthermore, even if tools for validating rules are available, there is still a considerable barrier in the form of the requirement to offer valid and acceptable IFC models as input, as well as GeoBIM models that still need to be prepared. The evaluation may be integrated with new contributions in future work. Other works and research, in addition to those described in this paper, should be combined into a more systematic body of work. Other factors that may enforce the transformation to a full DBP system, such as legal issues, should be considered.

Taking all of this into account, the next steps towards automating building code compliance checks in the AEC industry in the Kingdom of Bahrain should be aimed at filling the gaps listed below:

- Look at the AEC industry's infrastructure to see if it can completely switch from CAD tools to BIM tools to make all types of AE building drawings.
- Consider permitting building designs at the urban scale and integrating Geo-BIM techniques into any proposed DBP.
- Create a dynamic transition strategy while ensuring collaboration among all stakeholders, including contractors.
- Develop the necessary legal legislation to initiate the automated building code compliance checking system in the AEC industry.
- Develop a plan to train architects, engineers, and government employees in the departments responsible of BPs' works how to use software (like BIM, Geo-BIM, etc.) that supports digital transformation.
- Convert the local building regulations into a machine-readable format using the software chosen as the foundation for digitalizing the checking process.

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